Boying Tang

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EXPERIENCE

Software Engineer *at Aerware Technologies* April 2025 – Present

- Designed and implemented foundational backend APIs, powering the primary question generation and real-time presentation workflow supporting 100+ curriculum lessons.
- Built LLM-integrated backend workflows creating educational standards-aligned, personalized content using Python + OpenAI APIs, including schema enforcement, validation, and post-processing
- Modeled and maintained backend data schemas (DynamoDB) for lessons, questions, and metadata, supporting hundreds to thousands of persisted entities ensuring correctness, consistency, and efficient retrieval.
- Designed asynchronous generation pipelines with retry logic, guardrails, and structured output enforcement to improve reliability during high-load operations.
- Deployed backend services using AWS (ALB, S3, EC2) and Terraform, improving system reliability and making deployments reproducible across environments.

Software Engineer *at Amazon* August 2020 – March 2024

- Transitioned Chromium integration for Silk Browser from weekly manual scripts to semi-automated daily merges, reducing manual intervention to resolving conflicts roughly once per week while maintaining alignment with Chrome releases and ensuring code stability across a 25M+ line Android codebase.
- Architected and implemented automated testing frameworks, converting manual unit, integration, and performance tests into reproducible pipelines, reducing manual test time by ~20 minutes per release cycle while increasing test coverage and reliability.
- Optimized test parallelization and team ownership, breaking tests into team-specific pipelines to accelerate execution and improve ownership.
- Developed performance-signoff tooling, automating measurements of critical browser fluidity metrics on Android devices, enabling faster and more consistent verification before releases.

Student Developer *at NYU IT* May 2017 – January 2019, September 2019 – August 2020

- Designed caching strategies for microservices, improving system performance and reducing latency.
- Developed authentication and role management services for multiple departments using ASP.NET Core.
- Built and maintained full-stack solutions with React/Redux frontends and ASP.NET backends, enhancing API interactions and improving UX.
- Implemented RESTful API services, improving backend efficiency and scalability.

EDUCATION

New York University May 2020
Bachelor of Science in Computer Science GPA: 3.5
Dean's List Fall 2016 – Spring 2017, Fall 2018 – Spring 2019

SKILLS

- Backend: Python, Java, Node/Express, Go, API design, data modeling, schema design
- Distributed Systems: concurrency, async workflows, retries, idempotency, observability
- AWS: DynamoDB, S3 , ALB, CloudWatch
- DevOps: Terraform, Docker, CI/CD pipelines
- AI: OpenAI APIs, Cursor, LLM orchestration, schema enforcement, post-processing
- Testing: integration tests, backend validation, performance pipelines
- Tools: Git, Linux, automation frameworks